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HCI and deep time: toward deep time design thinking

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1. Introduction

To see objects not just through the lens of human agency but through the lives of nonhuman beings that both shape and are shaped by relationships and processes embodied in material forms is to invite stories – in fossilized bones, decaying tissue and living flesh. (Mitman et al., 2018, p. xi)

Several scholars within the field of human-computer interaction (HCI) have identified time and materiality as fundamental analytical facets of relevance to both the use and design of computer technologies (Berzowska et al., 2019; Odom et al., 2018). However, with some exceptions, there has been a (very reasonable) tendency to focus on relatively short-term interactions, delimited to situations of use or specific devices (although acknowledging, for example, issues of sustainability and temporal relationships and negotiations that take place across individuals, groups, and institutions).

As a contrasting example of particularly long-term interactions, humanistic research fields, such as media archeology, critical posthumanism, and the environmental humanities, have recently directed attention to the notion of *deep time* as a long-term perspective providing new analytical and ethical traction on both temporalities and materialities of media technologies (Fredengren, 2016; Mattern, 2015; Parikka, 2017; Taffel, 2016). Put simply, deep time refers to temporalities that include the fundamentally perdurable geological processes of the Earth – effectively considering the pace, rhythm, causalities, and materialities by which durable ecological changes occur. While such a perspective may at first seem difficult to grasp and remote to HCI and design, we take stock in environmental anthropologist Richard D.G. Irvine's argument that “deep time is not an abstract concept, but part of the phenomenal world impacting on people at the level of experience” (Irvine, 2014, p. 157). Paraphrasing Irvine, we thus suggest that one challenge for HCI and design is to find new ways of exploring the interactions between humans, technologies and geological temporalities. As far as the authors of this paper can tell, the notion of deep time has not been specifically theorized in the realm of HCI and design. While this in itself does not mean that there are no academic overlaps already, it also calls for an exploration of theoretical and practical intersections and synergies between deep time and related areas within HCI. As such, the main contribution of this paper is an initial outline of how HCI can take aspects of, what we may call ‘deep time design thinking’, into consideration. Having said that the suggestions and implications in this paper come from approaching deep time and related areas in HCI *analytically*. Our theoretical analysis of overlapping concepts and the proposed implications are not meant to be read as decisive evidence, but rather as analytical and practical tools that highlight how materiality (and thereby design) can be understood, and reflected upon, in terms of ‘deep’ temporalities. We consequently regard this paper as a first attempt at analyzing and articulating the potential of deep time design thinking in HCI.

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Before surveying already existing HCI research connected to temporality and materiality, an initial definition of deep time is in order. As mentioned, deep time is concerned with long-term geological processes, taking into account material developments and impacts over long durations of time. As such, deep time highlights how time also has a decisive tie to materiality, and thereby provides us with a perspective that is inherently tempo-material. Heringman (2015) describes deep time as essential to our understanding of the Anthropocene – the proposed geological epoch defined by significant human impact on Earth’s climate, ecology, and ecosystems – and that the concept offers us:

[...] a framework for confronting the carbon signature of the human species, but it also puts us in dialogue with the archaeological and geological signatures of ancestral species and geological processes themselves (p. 79)

For this paper, we will focus on deep time as it has been conceptualized by media theorist Jussi Parikka (2015a).¹ He proposes that a deep time in the media entails an interest in the very concrete materiality of technology. That is, by ‘excavating’ and tracing the raw materials used in production, and the relation media technologies have to geology – a discipline referred to by John Durham; Peters (2003) as the ‘science of time’ – an alternative, and media ecological, outlook can be highlighted (Fuller, 2005; Parikka, 2015b; Taffel, 2019).

In order to begin developing a notion of deep time design thinking, the paper follows a straightforward structure. It starts by surveying temporality-, materiality-, and sustainability-oriented strands of thinking already established within HCI, particularly emphasizing connections to deep time. It may seem counterintuitive for a paper on temporality to direct attention to materiality, however, as initially argued, and which will be more clearly explicated, there is an important connection between these two elements (particularly in relation to deep time). Next follows a similar survey, but drawing from the digital and environmental humanities, and focusing on deep time and the geological materialities of media technologies instead. Then follows a synthesis of the two theoretical surveys, outlining an initial proposal for ‘deep time design thinking’. Finally, the paper concludes with a discussion summarizing insights and phrasing future deep time-oriented research opportunities for HCI and design at large.

2. HCI, temporality and materiality

It seems important to begin by stressing that the ambition of this paper is not to ‘apply’ the concept of deep time to HCI – rather, deep time is a concept used to an increasing extent within the humanities, and HCI is already well equipped to enter these debates to the benefit of both HCI and the humanities (cf. Gross et al., 2014). As literature scholar David Farrier argues (Farrier, 2019), each discipline must consider deep time on its own terms, and reappraise its own boundaries and assumptions in the shadow of the Anthropocene. Following the presumption that time and matter are deeply connected, and have inherent ecological dimensions, previous research within HCI falls into three corresponding categories: temporality; materiality; and sustainability.

2.1. HCI and temporality

Naturally, research on HCI and time has been taking varying temporal scopes into consideration, ranging from specifically situated use to more political-economic issues stretched out in time. Toward the short-term end of the spectrum, Vallgård et al. (2015) explore ‘temporal form’ in interaction design, which they define as “the computational structure that enables and demands a temporal expression in the resulting design” (p. 1). As such, they focus on how the hard- and

¹It should be noted that there is another conceptualization of the deep time of media, which is concerned with exploring historically anomalous, but also temporally circular, media-technological development paths (Zielinski, 2006, 2019). Such research shows overlaps to design fiction as used within HCI, but is the topic of a different paper.

software of computational devices segment visualization and interaction, how this can be utilized in design, and interrogates what it means for interfacing with the human sensorium. Their conclusion is that such micro-temporal changes in interfaces enable a richer user experience than static ones.

In an ambition to go beyond delineated ‘processes’, or situations of use, in interaction design, Huang and Stolterman (2011) develop graph- and language-based representations of interaction attention over longer segments of time. The authors thereby extend the time-scope usually applied to user strategies in interaction design and open up to further explorations of long-termist interactions. In an effort to explore how the use of interactive systems is distributed over longer periods of time, Rattenbury et al. (2008) show how the ‘success’ of digital devices seems to be dependent on their temporal ‘plasticity’. That is, their ability to be used in parallel with other activities, supporting an experience of not switching between activities. According to the authors, the notion of plasticity captures how technologies have now been integrated into the heterogeneous rhythms of everyday life, where people are not always aware that, or how, they multitask. At the same time, constantly connected and ubiquitous digital technologies fulfill an affirmative function on the middle-class identity, where constant ‘busyness’ has become synonymous with being successful. Taking an even more extended perspective on time, Pschetz and Bastian (2018) build on humanist notions of temporality and argue that time emerges out of pluralist relations between social, economic, cultural and political forces, and that designers consequently “can help to create artefacts and systems that disclose this variety [of temporalities], also revealing the intricacies of temporal relationships and negotiations that take place across individuals, groups, and institutions” (p. 174). They further argue that such an effort could help to reveal intersections between mundane temporalities, as experienced in the everyday of users, and more durational temporalities of the natural world. While the authors do not specifically address deep time as part of such intersections, we find this appeal highly encouraging for our purpose of considering how materiality, as well as the relations to (posthuman) others, can be understood in terms of deep time.

When considering temporality and the increased awareness of the embeddedness of computer technologies in cultural and political contexts that the texts above highlight, another related important topic in HCI research is ‘slow technologies’ (Hallnäs & Redström, 2001; Odom et al., 2014; Skågeby, 2011) or ‘slow design’ (Grosse-Hering et al., 2013). The slow technology research agenda has, for example, emphasized designs that support mindfulness and reflexivity, designs that support multi-generational use, and designs that support critical consumption and less hectic lifestyles, among other things (Odom et al., 2012).

One of the things this brief survey of HCI and temporality illustrates is that, although there are various temporal spans at play, there is also a growing agenda to question values such as ‘efficiency’, ‘time-saving’, ‘speed’, and ‘busyness’ as inherently beneficial. The research on slow technology shows how slow and more long-term values are being put forward. There is also a certain tendency to incorporate the natural environment as a stakeholder of sorts when designing (cf. Smith et al., 2017). Both of these tendencies support a further engagement with deep time as a source of new insights into the complex reality of human-computer interaction.

2.2. *HCI and materiality*

A relatively short paper like this has little hope in summarizing all relevant literature on HCI and materialist thinking. Fortunately, there are previous efforts that we can lean on, which have partly aimed to summarize this body of literature, and partly aimed to contribute new theoretical additions. For example, in their exploration of overlaps between HCI and media-based theorizations of materiality, Gross et al. (2014) show how discussions and “references to materiality can be found throughout HCI, including its history and its subdomains”. The authors particularly identify three ‘active trends’ contributing to the theorization of materiality in HCI: tangible user interfaces, computational materiality, and craft, and relate these trends to an insightful discussion on the benefits and drawbacks of focusing on (material) medium specificities.

Several authors now discuss how interaction design is increasingly a question of addressing a composite between the computational and the material (Stolterman & Wiberg, 2020; Vallgård & Sokoler, 2010; Wiberg, 2017). While this body of work occasionally touches upon how materials can be a reminder of “cyclical, natural time” (Robles & Wiberg, 2011), as well as matters of ontology, their exploration is also mainly concerned with esthetics and compositional skills. In a similar way, Fuchsberger et al. (2014) consider different layers of materiality, concluding, much like Hayles (1999), that information always requires some kind of material form, and that digital materiality emerges in an interaction between different materialities.

Giaccardi and Karana (2015) develop a framework of ‘materials experience’, building ‘upwards’ from the experience of specific materials, toward interaction, and ultimately, practices. As such, the physicality of materials is somewhat downplayed in favor of embodiments, experiences, and performances.

Bergström et al. (2010) developed the interesting notion of ‘becoming materials’, acknowledging both technical (“climate, environmental, and human factors in potential contexts of use”) as well as esthetic (social and experiential) aspects. Their sophisticated argument tells us to consider materiality as increasingly super-sensory (and beyond human comprehension or perception), and design as a practice that can further our thinking about materials in the world at large. To us, this line of reasoning further supports the idea of developing deep time design thinking.

HCI research has, of course, also taken interest in ‘ecological’ aspects of interactive artifacts (Blevins et al., 2015; Forlizzi, 2008; Jung et al., 2008). In these accounts, the theoretical vantage point mainly relies on studies of complex information systems as *environments* – and thus as interactions between media technologies, techniques and human feelings, thoughts, values, and social behaviors. In this conceptualization of media as environments, it is mainly humans (and their feelings, thoughts, values, and social behaviors) that remain at the center of analysis, which also resonates with one strand of ecological thinking within media studies (Cali, 2017; Strate, 2017). However, in a different version of media ecology, leaning more on theorists such as Bateson, Guattari, Hayles, Latour, Kittler, Fuller and Hörl, environment and ecology is not an equivalent concept. Instead, these accounts are often skeptical toward anthropocentrism and technological solutionism, and instead emphasize materialist theories which treat several actants, not only humans, as part of a broader and deeper conceptualization of (media) ecology. Here, the world is not just a passive preexisting stable environment that is subjected to various technological or human actions and solutions. Rather, there is a fundamental co-development of humans, machines, materials, natures, and cultures, where specific consequences emerge from complex relations, or intra-actions, between many different agencies, and across longer stretches of time.

Having said that, the relatively firm focus on ‘material interactions’ in HCI is not surprising, and represents an important and innovative strand of research – “informed by an exploration of emerging materials and technologies, and primarily grounded in interaction design research and practice”, as Giaccardi and Karana puts it (p. 2447). Insights into how materialities and immaterialities, hardware and software, intra-act and co-develop (cf. Dourish, 2017) are significant in developing interaction design in both research and practice. Nevertheless, while these accounts, as we have seen, address materiality in terms of physical, computational and cultural aspects of interaction, they also seem to retain two aspects that ‘deep time’ could elaborate: a human, and to some extent technology, -centric perspective; and a temporal orientation which focuses on recurring conventional styles in preceding technological forms, or what Murray (2011) refers to as legacy media, rather than an elongated focus on ecological and geological time. So, even though materials and materiality are considered, deep time opens up to issues around geopolitical temporality – including both archeology and the making of futures.

2.3. HCI and sustainability

As a part of its material turn, recent HCI research has also paid much attention to issues of sustainability (Bates, Thomas & Remy, 2017; DiSalvo et al., 2010; Dourish, 2010; Raghavan &

Pargman, 2017) – issues that are connected to the concept of deep time. Two of the more elaborate and insightful accounts are provided by Light et al. (2017) and Knowles et al. (2018). Firstly, in their paper *Design for Existential Crisis in the Anthropocene Age*, Light et al. propose four ‘moods’ that can encourage “meaning, purpose and fulfillment in difficult, unstable and rapidly changing times”. Their initial proposal is to *design for attention*. More specifically, they refer to attention toward our larger ecological surroundings, which can challenge anthropocentrism and promote awareness, recognition and empathy with more-than-human circumstances. Next, with the idea of *being different*, the authors highlight temporal dimensions of both ecological and technological nature, with a focus on preparing for the future in creative and flexible ways. By designing for *being in it together*, the authors emphasize an orientation toward more holistic commonalities and synergies and “to design with a sense of ecology and a need to balance competing interests in their ever-changing entanglement”. Finally, using the notion of *avoiding bovine design*, Light et al. discusses how design may work in conforming ways, which, in the worst case, delimit agencies and potentials.

Secondly, Knowles et al., leaning on Naomi Klein’s book *This Changes Everything: Capitalism vs The Climate*, outline nine areas where sustainable HCI (SHCI) research have made, and can continue to make, important contributions: systems thinking; computing within ecological limits; future proofing; confronting the economy; supporting local infrastructures and communities; advocating social justice; inspiring with future visions; reimagining (digital) civics; and embracing and accounting for the political. In brief, systems thinking implies that “[...] sustainability interventions must be conceived of and evaluated between different scales, over longer timeframes, and across a greater diversity of stakeholders”. Computing within ecological limits and future proofing acknowledges that natural resources are finite, and that such restrictions should guide design research problems and solutions. In confronting the economy, the authors suggest that SHCI research could challenge and interrogate economic operations that may be taken for granted. This relates to the issue of supporting local communities and infrastructures, where a focus on decentralizing resources and promoting communal values. Advocating social justice entails awareness of how media technologies can reduce or perpetuate social inequalities, in terms of class, race, gender, corporeality, etc. With the term inspiring future visions, the authors discuss how design fiction can help in promoting alternative, more sustainable, conceptualizations and foresights of societies and the role that media technologies have within them. The idea of reimagining (digital) civics raises the issue of how HCI can enable impactful civic participation, both within the citizenry and within the research community. Finally, in embracing and accounting for the political, the authors question and confront the ideas of whether technology can ever be apolitical, and that they are merely neutral tools that can be used in ‘good’ or ‘bad’ ways. The authors predict that these areas, and the way they frame problems and solutions, will generate some controversy among the (S)HCI community, but also argue that a move from thinking to action requires an accountable ethical agenda that acknowledges systemic real-world problems.

While our survey of HCI research on temporality, materiality and sustainability has shown that there is significant ground covered, it also shows opportunities to advance the current understanding and conceptualization of temporality in HCI. Firstly, HCI research could extend its temporal scope to acknowledge the interplay between profoundly long-term geological and ecological time-scales and contemporary technology use and design. Secondly, to go beyond an anthropocentric and to some extent technology-centric perspective on technology use and design. While this may seem to question the very notion of *human-computer* interaction, it also puts technology in direct connection with our atmosphere, biosphere, hydrosphere and geosphere. Thirdly, for sustainable technologies to be truly sustainable, we cannot continue to make geo-political separations of the here and now from the non-negotiable long-term implications for our planet and future generations of beings. As a consequence, it seems important to try to develop a notion of design thinking that connects these insights and HCI, which is the focus of the next section.

3. Connecting HCI to deep time

As mentioned, deep time is concerned with geological timescales, spanning up to 4.5 billion years. As such, its relevance for experienced human time or its contemporary cultural significance, not only in relation to HCI, can be hard to fathom (Irvine, 2020; McPhee, 1981). In an insightful attempt to make deep time and its implications on our mundane reality more graspable, Cervato and Frodeman (2012) develop a three-part framework. Firstly, they ask us to consider how human societies are dependent on a wide range of natural materials which “have formed over geological time spans under conditions that are difficult or impossible to reproduce” (p. 27). A part from constituting energy supplies and basic life necessities, many of these materials also form the constructional foundation for designing computer technologies. However, the cost of extracting such materials is often externalized, and current economic models often suffers from extreme short-sightedness. As such, it raises the question of how HCI can take geological time on board in order to counteract much of current technology development’s shallow time perspective. Secondly, by working across disciplinary borders, HCI – as a discipline often making explicit connections between its research and its usefulness to society – can make research and design decisions that are placed within a framework of both the geologic past and the geologic future, which also have a tangible impact on human decision-making. Thirdly, Cervato and Frodeman argue that a cultural model driven by the acquisition of possessions (e.g., *Homo economicus*) is reaching its (eco)logical end. In this sense, HCI could help in promoting a revisit to less destructive models of consumption and disposal, as well as more ‘existentially sustainable’ modes of being and becoming.

Recently, the concept of deep time has been adopted by media materialist and digital humanities research, emphasizing ecological, and even geological, strata of media technologies and their design (Fuller, 2005; Parikka, 2016; Taffel, 2019). The notion of ‘deep time’ – a perspective on the very primal matter (or ‘staple’ in media theorist Harold Innis’ terms) of technological development – is acknowledging and accounting for the geological foundation existing in both contemporary and future material politics of media technologies. In this sense, deep time is extending our view of what parts that actually make up a joint cognitive system (Hollnagel & Woods, 2005) or a human-technology relation (Rosenberger & Verbeek, 2015). While previously focusing mainly on various forms of human-technology interactions, a deep time perspective could be said to add a geological component to the equation: human-technology-geology. As Parikka (2017, p. 177) puts it:

- (1) Geology refers to the affordances that enable digital media to exist as a materially complex and political-economically mediated realm of production and process: a metallic materiality that links the earth to the media-technological
- (2) Temporalities such as deep time are understood in this alternative account as concretely linked to the non-human earth times of decay and renewal, but also to the current Anthropocene and the obscenities of the ecocrisis

This means that, when considering deep time, the interaction-paradigm arguably extends beyond immediate user experiences, and highlights a different kind of tempo-materiality than is usually emphasized within HCI. In the words of Parikka again: “Media history conflates with earth history; the geological materials of metals and chemicals gets deterritorialized from their strata and reterritorialized in machines that define our technical media culture.” (Parikka, 2015b, p. 35). At the most fundamental level, this is played out in the mining of so-called conflict minerals – many of them vital to IT products that are sold globally. Such mining has been, and arguably continues to be, at the source of many ensuing complications: environmental pollution, human rights violations, health problems, corruption and war, for example. That is, as valuable natural resources, the minerals are extracted and traded, often under violent conditions and illegally, in order to fund paramilitary operations. As such, the ethics of extracting tin, tantalum, gold, tungsten and cobalt comes to the fore, and new supranational regulations are coming into place to help stem the trade in these conflict

resources. When connecting these issues to HCI, it highlights the need to consider the materials that make up contemporary technologies, not only as parts of the deep time of the earth as it plays out now but also in terms of future remains and future fossils (Farrier, 2020; Mitman et al., 2018). What kind of fossil traces will HCI and (designed) technologies leave behind for future generations to find?

The role of technology in theorizations of deep time is one where an ‘originary technicity’ or a fundamental ‘technological condition’ is at the center. That is, humans are not only designers of technology but also, in themselves, irrevocably constituted by and through technology (Bradley, 2011; Siegert, 2015). The extension of this idea into the Anthropocene also renders the human relation to ecology and geology as inherently technical, and as something:

[...] that constantly merges human and non-human actors, that operates before the difference between subject and object, that is endlessly prosthetic and supplementary, that is immanent rather than transcendental, and that is to an unheard-of degree distributed and indeed ecotechnological (Hörl, 2015)

The fact that this technological condition, where (media) technologies fundamentally transform subjectivities, agencies, and ontological boundaries, is now starting to operate through ubiquitous computational infrastructures and devices, and thereby only strengthening its grip on cultural and material realities in a wide (even ecological and geological) sense, would suggest that HCI is, can, and should be a core discipline in tackling such issues. Indeed, philosopher David Wood claims that deep time, and its connection to new materialist thinking, can provide new ways of thinking about agency, responsibility, and community in the Anthropocene:

We can then feed back a new ontology—a new imaginary, a much richer understanding of collaborative agency—into our socionatural practices. If the climate crisis is at least partially a product of translation of an imaginary of human sovereignty and domination over nature into a free market economy allergic to ecological constraint, new materialism’s novel way forward should work at this level. It is not just a new vision (say, of harmony), nor just a new attitude. What this new materialism offers are new discursive practices, new ways of analyzing problems and situations, and new ways of understanding the shape of our engagement. (Wood, 2019, p. 103)

So, while deep time may, at first, have seemed a bit remote from HCI and interaction design, there is a clear connection in the very midst of contemporary digital culture. HCI already has a rich understanding of the complexities of agency, responsibility and community in human-technology relations. This means that HCI is a discipline where meaningful and ethical action can really make a profound difference: “Not only do contemporary digital media physically connect us to geological time and globalized space, they also present the agglomeration of past life in the service of a present life that is actively remolding the potential for future life via their contribution to contemporary ecological crises.” (Taffel, 2016, p. 370). As HCI has moved from users toward larger (eco)systems and media ecologies, a linking to deep time can broaden this momentum even further, and connect it to both deep time, and post- or more-than-human ethics. Specifically, this could enable HCI to address materiality beyond its “specific abilities to support functionality and task completion”, “how materials contribute to the aesthetic qualities of a design” (Gross et al., 2014), or partial and delimited ‘artifact ecologies’ (Bødker, 2015), and tackle general ecological, and even geological, interaction as well.

At the same time, there seems to be a slight difference in terms of the viability of (technological) solutionism. From the HCI strand, there seems, perhaps for readily apparent reasons, to exist a stronger belief that HCI has the capacity to change and challenge current development trajectories, and that new computer technologies can possess the power to alter fundamental infrastructures. From the humanist strand, there is arguably a larger degree of skepticism and pessimism (cf. Allenby, 2007; Wood, 2019). As philosopher and feminist theorist Elizabeth Grosz puts it in an interview with Heather Davis and Etienne Turpin:

I don’t want to suggest, however, that the age of man can be overcome by man. This is perhaps the very same arrogance that produced the Anthropocene, the human tendency to believe in its own power, its problem-solving capacities, even as it has undertaken ruinous activity that imperils other forms of life. It is only hubris

that leads us to believe that if the human has polluted and nearly destroyed vast environments, it can somehow restore what it has destroyed. It may be that we have not only summoned up a new architecture, new arts, new ways of living, but new forms of self-destruction, new ways of overcoming the human, new ends for architecture. (Davis & Turpin, 2013, p. 132)

On a similar note, and specifically relating to the naturally, socially, culturally, economically and politically integrated complex information infrastructures we now live in, Allenby concludes that:

The world has become media, but not in the oversimplistic sense of being completely anthropogenic. Rather, it has evolved to levels of complexity and constant change that combine physical and information systems in ways that we as a species have not had to contemplate, or manage, or, least of all, design intentionally, and we certainly don't understand. (Allenby, 2007)

A similar line of thinking is demonstrated by Mattern (2015) who argues that a focus on deep time can also entail a focus on *infrastructure* prior to the introduction of modern media. In a proposal for a number of lessons that media and design studies can take from archeology and history of architecture, she outlines how a deep time of infrastructure “can encourage us to reflect critically on how we construct our media” (p. 95). Essentially, Mattern argues that infrastructures can be seen as layered, illustrating how their different temporalities reside, co-exist, and take on new cultural meanings as new infrastructures are built on top of others – something which has also been visible within HCI (Pipek & Wulf, 2009). Thus, considering *how* new technologies enter into such intertwined temporalities and residual information infrastructures, that may have long durations and histories, becomes crucial – specifically in terms of technologies having real material and political effects across broader scales of time. Importantly, such infrastructures impact not only on the lives of *people* in the *present* moment but on general ecologies (and not just artifact ecologies), and have done so for a long time, and will continue to do so for a long time. While these accounts can be read to suggest that the complexity of ‘medianatures’ is overwhelming, there is really no alternative but to keep exploring these wicked problems and systemic intricacies, and in this endeavor, deep time can provide us with new tempo-material sensibilities for thinking about HCI and design in the Anthropocene.

4. Toward deep time design thinking

Based on the two theoretical surveys, this section will synthesize some initial insights that can be used for developing, what we will call, ‘deep time design thinking’. We chose this epithet as a way to highlight an ambition to move beyond human-centeredness, rapid problem-solving, exclusion of stakeholders and conceptualizations without real impact (for which ‘design thinking’ has been critiqued) and instead toward ‘design thinking with depth’ (Kolko, 2018). We do not propose that these insights are coherent processes or strategies in themselves, but rather think of them as initial mind-sets, or sensibilities, that can help designers and HCI researchers to address different classes and timescales of wicked problems. Nor do we think that our proposals are the only possible sensibilities that are of use when considering deep time in HCI and design.

4.1. A multiple timescale sensibility

To consider the interplay between the mundanity of everyday life and geological timescales (and everything in-between) is difficult, but it is nevertheless necessary when reflecting on the long-term impacts of computer technologies. HCI could thus scrutinize, what David Farrier calls, the “ethical proximity between the most fleeting act in our present and planet-shaping effects that will play out over millennia” (p. 2) in relation to interactive technologies. This includes acknowledging that humans are inseparable from other planet-shaping powers, and working toward not disconnecting HCI and design activities from the long-term geological and ecological rhythms on which they depend (Irvine, 2020). Such an oscillation between the short-term timescales that are immediately

sensible to users, and the deeper histories and futures of the larger environment, could help both designers and users reflect upon long-term consequences of lifestyles and materialities. Further, interplays with infrastructural time, such as everyday activities, work schedules, expectations of communication speed, the tempo of consumerism, the pulse of urban technologies, and so on, provides a meso-level perspective on time and technology. Finally, to also consider the enduring traces that will persist into deep futures. What future remains and future fossils could be the result of specific designs?

The Anthropocenic challenges that we are facing could be seen as an opportunity for designers and HCI researchers to approach wicked problems from multiple and oscillating timescales. Considering the central position that HCI holds in shaping interactions in a technology-saturated society, it also bears a potent agency to engage with, and even foster, an awareness of multiple timescale contingencies in both research, design, and its more encompassing societal consequences.

4.2. A post-humanist sensibility

Notably, research within both HCI and the digital and environmental humanities are beginning to question the centrality of the human in our larger ecologies. Recent scholarship emphasizes how both our entanglement with the natural environment and our relationship to technologies is a fundamental precondition rather than a discrete externality. One of the aspects that deep time highlights is that both humans and technological objects are entangled in deep, geological and ecological, histories and futures. Thus, one sensibility to develop is to go beyond an anthropocentric and to some extent techno-solutionist perspective on technology use and design. How can we, through our research and design activities, cultivate collaborative rather than exploitative relations with other species and the environment over time?

As mentioned, HCI has begun to question ‘strict’ user- and human-centered design, and acknowledged, for example, new co-active partner technologies (Giaccardi & Redström, 2020; Skågeby, 2018) or animal-computer interaction (Mancini, 2011). What deep time does, however, is to expand this analysis even further, to include not only algorithms, social robots, animals and other performative actants in the technology design space, but also geological materialities and temporalities. In this sense, the research and design agenda expand temporally and materially. As Light et al. suggests (Light et al., 2017), maybe HCI is not the best acronym anymore, as we begin to account for the flows of resources, both human and non-human, that converge in our designs (the *what*, *when* and *where* of humans and computing). Thus, complementing anthropocentric usability studies with media-materialist and post-human deep time-oriented studies seems fruitful.

4.3. An anti-presentist and eco-political sensibility

Interestingly, several of the surveyed studies emphasize critiques toward the current moment – what social anthropologist Irvine (2020) has referred to as ‘the problem with presentism’, where time horizons collapse into an omnipresent ‘here and now’. Much like anthropology, HCI occupies an awkward position in relation to presentism. Even when going beyond response times and task analysis, user studies and design are often time- and place-situated practices, potentially abstracting certain moments and solutions from their wider temporal contexts. Apart from their ‘inherent strive’ to improve processing speed, computer technologies are also consumer goods, making them suffer from market logics, where timing is often correlated to a maximization of short-term profit. However, the present is also our most important moment, where change can be enacted (and designed). It is only when it is circumventing deep time sensibilities, sidestepping accumulated interdisciplinary knowledge, or ignoring cumulative impacts on ecologies, that presentism becomes a potential hazard. In essence, we should avoid reducing human-computer interactions to relationships in the present only. Such an anti-presentist sensibility can, naturally, take different forms.

As previously touched upon, in HCI, ‘emerging materials’ is of significant interest, which means that there are also reasons for designers to actively consider how the preceding (mineral) materiality plays into the subsequent process of materializing products and media technologies. Importantly, this consideration should also extend to the end of the product life cycle, when processes and phases of obsolescence and waste are introduced. While planned obsolescence is a design practice detrimental to the ethics of deep time, it is nevertheless a significant part of how consumer society pushes a continual upgrading, and at the same time a continuous thrashing of technology which is still functional (Pope, 2017). When technologies are discarded at an increasing rate, they contribute to an increasing electronic waste pollution problem (Parikka, 2012). A problem that is perhaps particularly tangible in developing countries (to where e-waste management is often relocated), due to the economic logic of globalization where places of production (and waste) are separated from places of profit. The short-termist economical and time-saving benefits are located in one place and the long-termist geological and ecological drawbacks are located in another. This geographical separation of effects of technology often hides this “dark side” of technology (as it happens elsewhere) and instead over-emphasizes their local and current (often time-saving) benefits. E-waste becomes a serious ecological problem in some areas (‘sacrifice zones’), where the pollution of other natural resources becomes an additional risk (e.g., air, water and soil). In combination with a lack of legislation and regulation, e-waste thus ends up in, sometimes massive, landfills chock-full of hazardous metals. It is further not uncommon that e-waste is handled by (literally as well as legally) unprotected children in order to extract metals that are of some value (e.g., aluminum and copper). A lot of e-waste is also burned as it is, releasing toxic airborne fumes and particles – eventually ending up in human bodies and natural environments (targeted already vulnerable bodies and places more than the already privileged). Of course, metals are not the only cause for media geological concerns: plastics also create “[...] conversations between geological and industrial temporalities, foregrounding contemporaneous ecological crises and the material politics surrounding entanglements of media, plastics, and ecosystems” (Taffel, 2016, p. 357). As such, from a media geological point of view, storage formats, technologies, and infrastructures all demonstrate a connection between geology, and an exploitation of natural, as well as human, resources (Devine, 2019). Thus, conflict mineral mining, planned obsolescence, and e-waste become underpinning ethical problems that the notion of deep time reveals (Cubitt, 2014), where the immediate ecological consequences are handled locally (away from the site of consumption and use), but the overall consequences need to be taken seriously, including within design processes. As such, it is obvious that artifacts have agency, in an ecological sense, even after the technologies are defunct and discarded. Plastics, salts, silicone, and metals are becoming new focal points for cultural analyses, raising questions about the design as an inscription of material agency – what does this technology *do* – and when does it perform these actions – not just when computationally operational, but also when (still) ecologically operational? As such, media technologies are often based on a bracketing of deep time. Materials that have been created geologically over long time spans are used and, in fusion with other materials, processed into a product or energy which is used for a short period of time (and then becomes rubbish). The metals (and energies) cannot be equally quickly returned to nature in a way that recycles what has been. The re-creation, if even possible, instead takes hundreds, or even thousands, of years. In this way, the conditions of the present are created by using what has been created over long periods of time. This ‘loan’ from history is then repaid in an altered state. The present (where time is money) is thereby robbing both the past and the future of its ecological sustainability.

Further, an anti-presentist sensibility should also raise the question of what counts as successful design and for whom. A resignification of success (Halberstam, 2011) may be in order to question values that reoccur as taken-for-granted paradigms in design, business, and society at large. Halberstam suggests that we look for alternatives to ‘over-established’ understandings of success that perpetuate and mediate social, political and economic interaction. While we should still avoid ‘bovine design’ (Light et al., 2017) there could nevertheless be reason to revisit obsolete media forms in order to pick up on

design and use values that can be reanimated. The slow media (Rauch, 2018) and slow technology movements (Hallnäs & Redström, 2001; Odom et al., 2014), as well as media archeological practices (Hertz & Parikka, 2012), provide new temporal frames for, both reviving and projecting, problematizations and solutions for HCI. Technological design, both presentist and speculative, has a role to play here in empowering individuals, groups, societies, and even infrastructures, in inspiring, nudging, and promoting meaningful agency, or even actively hindering (‘undesigning’) certain actions. Even more so now, when opting out of pervasive digital infrastructures (that are often too vast to grasp or experience in themselves) is not a viable alternative for most people (and even less so for other agents).

5. Conclusion

The current ecological crisis presents us with wicked problems of an unprecedented scale. Temporality and materiality are important analytical concepts for tackling such problems. In particular, this paper argues that deep time, as a concept addressing geological tempo-materialities, can be invoked to develop new sensibilities in HCI research and design. That is, the ways in which we address (or do not address) the ecological crisis through the technologies we design is paramount. Designs produce effects, and as such have the potential to change attitudes, behaviors, practices, and ultimately control and governance. An increasing body of work is showing that the ‘political agency’ of digital technologies should not be underestimated. As such, the scope of HCI is also expanding. The (perhaps never stable) concepts of user (human), computer, and interaction have been diversifying as ICTs get increasingly ubiquitous and infrastructural. The integral position of media technological systems in modern societies puts the design of them in a vital position to enact change and awareness. In an era where we are witnessing how increasing demands on natural resources are having an impact on ecologies, large and small, the question of how HCI can, and should, respond is highlighted. This paper has presented a first attempt at connecting the concept of deep time to related areas of temporality, materiality and sustainability within HCI research. As such, the paper has made three contributions. It has argued that time and temporality cannot be separated from matter and materiality. Deep time illustrates how time can be regarded as vertical (structured as sediments and layers) rather than horizontal (structured as teleological progression). By connecting both humans and computers interaction to deeper scopes of vertical geological time, new tempo-material sensibilities can be highlighted for HCI research and design. As such, the paper also shows that research on deep time can help further expand the HCI research agenda on, for example, slow technologies and sustainable HCI. These sub-disciplines represent important and timely (in several senses) areas of research, where deep time could help to raise even further questions and solutions. Finally, in a more applied sense, the paper also proposes deep time design thinking as an area for consideration in HCI research and design, and specifically suggests three types of sensibilities that connect and ground the potentially enigmatic concept of deep time to HCI and design. We hope that future research will take up our call and provide designers and HCI researchers with more refined sensibilities of deep time, and also study if and how these sensibilities are then applied when designing.

Declaration of interest statement

No potential conflict of interest was reported by the author(s).

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